KNUS–OJANGUREN'S RECOLLEMENT (3.2) FOR TWISTED MODULES

Abstract. We rewrite Lemma 3.2 of Knus–Ojanguren (LNM 844, p. 224) for twisted modules and list every ingredient it uses. One ingredient is not stated by the authors: that some multiple of a class of positive rank in K_0 is the class of a projective module. This fact is not elementary. For the use we have in mind (gluing over a Zariski cover on whose pieces the gerbe is trivial) we give a different argument that avoids it. The argument needs only Lombardi–Quitté X.5.5 ($\widetilde{K}_0$ is nil) and XVI.3.4 (Quillen's patching lemma), together with elementary manipulations of K_0.

1. Setting

Let A be a commutative ring and G a $\mathbb{G}_m$-gerbe over $\operatorname{Spec} A$. A *twisted module* on an open U is a family V of finitely generated projective modules over $G|_U$ on which each loop $a \in R^\times$ acts as multiplication by a (weight 1). Weight-k modules are defined in the same way. A weight-0 family descends to an ordinary module on U. In particular, for twisted V, W the modules $\operatorname{Hom}(V,W)$ and $\operatorname{End} V$ are ordinary. Twisted modules are glued along $D(a) \cup D(b)$ like any other family, since $D(a) \vee D(b)$ is the pushout of $D(a) \leftarrow D(ab) \rightarrow D(b)$.

Torsion. The class of G has order dividing n if and only if there is a weight-n twisted module Λ of rank one on all of $\operatorname{Spec} A$. We fix Λ, and for a twisted V we set

$$P(V) = V^{\otimes n} \otimes \Lambda^{-1},$$

an ordinary module of rank $(\operatorname{rk} V)^n$. One has $P(V \otimes F) = P(V) \otimes F^{\otimes n}$ for F ordinary.

K_0. We write $K = K_0(U) \otimes \mathbb{Q}$. For F of constant rank $N \geq 1$ put $\rho(F) = [F]/N \in K$. A twisted module V of constant positive rank is *good* if $\rho(P(V)) = 1$ in K, i.e. some $P(V)^k$ is stably free.

2. KO 3.2 rewritten

KO 3.2 states: if $c \in H^2(A)$ is torsion and $c|_{A_f}$, $c|_{A_g}$ are represented by Azumaya algebras, then so is c. In twisted form: given twisted V on A_f and W on A_g, with ranks r and s, produce a twisted module on A.

The datum. On A_{fg} put $H = \operatorname{Hom}(V,W)$ (rank rs) and $E = \operatorname{End} V$ (rank r^2). The canonical isomorphism

$$V \otimes H \;\cong\; V \otimes V^\vee \otimes W \;\cong\; W \otimes E \tag{1}$$

is KO's datum (A, B, P, Q, α) with $P = H$ and $Q = E$. KO first twist an arbitrary datum by a line bundle $I \in \operatorname{Pic}(A_{fg})$ so that its class is c, and then use the injectivity of $\operatorname{Br}(A_f, A_g, A_{fg}) \rightarrow H^2(A)$ (their Theorem 2.11) to see that the n-th power of the datum is trivial. Here (1) is already a datum for G itself, so neither step is needed. Its n-th power is the isomorphism

$$P(V) \otimes H^{\otimes n} \;\cong\; P(W) \otimes E^{\otimes n}, \tag{2}$$

which is KO's $w : M \otimes P^{(n)} \cong N \otimes Q^{(n)}$ with $M = P(V)$ and $N = P(W)$.

KO's computation. Let $z \in K$ be the n-th root of $\rho(P(V))^{-1}\rho(H)^{-n}$. Then $z\rho(H)$ is the n-th root of $\rho(P(V))^{-1}$, which comes from A_f, and $z\rho(E)$ is the n-th root of $\rho(P(W))^{-1}$,

Date: November 2, 2026.

which comes from A_g. KO then pick an integer h and modules X on A_{fg}, P' on A_f, Q' on A_g with $hz = [X]$, $hz[H] = [P']$, $hz[E] = [Q']$. Applying Bass's cancellation theorem to $V \otimes X \otimes H \cong W \otimes X \otimes E$ gives an isomorphism $V \otimes P'^m \cong W \otimes Q'^m$ for large m, and gluing gives the result.

Ingredients.

(1) n-th roots in $1 + \widetilde{K}_0 \otimes \mathbb{Q}$, existence and uniqueness. Uniqueness is used, without being stated, to identify $z\rho(H)$ with the restriction of a root computed on A_f. Both follow from LQ X.5.5 ($\widetilde{K}_0$ is the nilradical of K_0), with the root given by a finite binomial series.

(2) *Effectiveness:* for $x \in K_0 \otimes \mathbb{Q}$ of positive rank, hx is the class of a projective module for suitable h. KO quote Bass for this in one sentence. It is not elementary. If L is a line bundle with $\ell = [L] - 1$, $\ell^3 = 0$ and $1, \ell, \ell^2$ linearly independent in $K_0 \otimes \mathbb{Q}$, then $8[L]^{1/2} = 3 + 6[L] - [L^2]$, and no multiple of it is a combination with nonnegative coefficients of powers $[L^k]$, $k \in \mathbb{Z}$: such a combination would be an $\mathbb{Z}$-valued distribution with mean $1/2$ and variance 0. So a non-split bundle has to be constructed. The classical proof reduces to a finitely generated $\mathbb{Z}$-algebra and applies Serre's splitting theorem; constructively this is LQ XIV.3.4 together with the dimension bounds of LQ XIII and XIV.

(3) Cancellation. Only the weak form below is used, and it is elementary (Lemma 3.2). Moreover X may stay virtual, because (1) gives $[V][X][H] = [W][X][E]$ for every $X \in K_0(A_{fg})$.

(4) Constant ranks ("sans restriction de la généralité"). In the use below all ranks are constant by construction.

Of these, only (2) is hard. The rest of the note shows that it is not needed for the use we have in mind.

3. Gluing good twisted modules

Lemma 3.1. *Let F have constant rank $N \geq 1$. If $\rho(F)^n = 1$ in K, then $\rho(F) = 1$.*

Proof. Put $\mu = [F] - N$. It is nilpotent by LQ X.5.5. From $k([F]^n - N^n) = 0$ we get $k\mu(c + \mu q) = 0$ with $c = nN^{n-1}$ and some q. Then $kc^t\mu = 0$ once $\mu^t = 0$, by expanding $c^t = ((c + \mu q) - \mu q)^t$. □

Lemma 3.2 (weak cancellation)**.** *Let X, Y, T be twisted modules with $X \oplus T \cong Y \oplus T$, where X and Y have positive rank. Then $X^{2m} \cong Y^{2m}$ for some m.*

Proof. First, T is a direct summand of some X^m. The evaluation map $X \otimes \operatorname{Hom}(X, T) \to T$ is surjective, since this can be checked locally, where $X = L \otimes F$ with F of positive rank. It splits, because $\operatorname{Hom}(T, X \otimes \operatorname{Hom}(X, T)) \to \operatorname{End} T$ is a surjection of ordinary finitely generated projective modules. Finally $\operatorname{Hom}(X, T)$ is a summand of some O^m. Similarly, T is a summand of Y^m.

Write $X^m \cong T \oplus T'$. Then $X^{m+1} \cong Y \oplus T \oplus T' \cong Y \oplus X^m$, and by induction $X^{2m} \cong Y^m \oplus X^m$. In the same way $Y^{2m} \cong X^m \oplus Y^m$. □

Proposition 3.3 (gluing good modules)**.** *Let $a, b \in A$ be comaximal. If V on A_a and W on A_b are good, of ranks r and s, then for some m there is an isomorphism $V^{ms} \cong W^{mr}$ on A_{ab}. Gluing gives a twisted module V' on A whose restrictions to A_a and A_b are good.*

Proof. From $H^{\otimes n} \cong P(V)^\vee \otimes P(W)$ we get $\rho(H)^n = 1$. The dual of a stably free module is stably free, so $\rho(P(V)^\vee) = 1$. Lemma 3.1 then gives $\rho(H) = 1$, and likewise $\rho(E) = 1$. Hence

$k([H]-rs)=0=k([E]-r^2)$ for some k. Tensoring the corresponding stable isomorphisms with V and with W and using (1) gives $V^{krs}\oplus T\cong W^{kr^2}\oplus T$ with $T=V^j\oplus W^{j'}$. Lemma 3.2 gives $V^{2m'krs}\cong W^{2m'kr^2}$; take $m=2m'kr$. Goodness of the restrictions holds because $P(V^e)=P(V)\otimes O^{e^n}$. □

The glued V' need not be good on A. Its defect lies in the kernel of restriction, which we now describe.

Gluing matrices. For $g\in \mathrm{GL}_M(A_{ab})$ let P_g be the module glued from A_a^M and A_b^M along g. Then $P_{\beta g\alpha}\cong P_g$ for $\beta\in\mathrm{GL}_M(A_b)$ and $\alpha\in\mathrm{GL}_M(A_a)$, and $P_g\otimes P_h=P_{g\otimes h}$.

Lemma 3.4 (homotopy)**.** *Let $\varepsilon(T)\in\mathrm{GL}_M(A_{ab}[T])$ with $\varepsilon(0)=I$. Then $P_{g\varepsilon(1)}\cong P_g$.*

Proof. Apply LQ XVI.3.4 to ε over $(A_a)_b$, and to $\phi=g\varepsilon g^{-1}$ over $(A_b)_a$. If $U(T,Y)$ works for s in XVI.3.4, then $U(T,s'Y)$ works for ss', so one N serves both cases. For large N this gives $\varepsilon(T+b^NU)\varepsilon(T)^{-1}\in\mathrm{GL}_M(A_a[T,U])$ and $\phi(T+a^NU)\phi(T)^{-1}\in\mathrm{GL}_M(A_b[T,U])$. Write $1=a^Nx+b^Ny$. Then

$$g\varepsilon(1)=\phi(1)g=\left[\phi(b^Ny+a^Nx)\phi(b^Ny)^{-1}\right]g\,\varepsilon(b^Ny),$$

where the bracket is the specialisation $T=b^Ny$, $U=x$ of the second matrix, so it lies in $\mathrm{GL}_M(A_b)$, and $\varepsilon(b^Ny)$ is the specialisation $T=0$, $U=y$ of the first, so it lies in $\mathrm{GL}_M(A_a)$; this uses $\varepsilon(0)=I$. □

Lemma 3.5. *Let $g,h\in\mathrm{GL}_M(A_{ab})$ and $g'\in\mathrm{GL}_{M'}(A_{ab})$.*

1. *$[P_g]+[P_h]=[P_{gh}]+M$. Hence $[P_{g^j}]=j[P_g]-(j-1)M$ for all $j\in\mathbb{Z}$.*
2. *$([P_g]-M)([P_{g'}]-M')=0$.*
3. *If $x\in K_0(A)$ restricts to 0 in $K_0(A_a)$ and in $K_0(A_b)$, then $x=[P_g]-M$ for some g.*

Proof. (1) $P_g\oplus P_h=P_{\mathrm{diag}(g,h)}$ and $\mathrm{diag}(g,h)=\mathrm{diag}(gh,1)\,\mathrm{diag}(h^{-1},h)$. Whitehead's identity $\mathrm{diag}(h^{-1},h)=u(h^{-1})\,l(-h)\,u(h^{-1})\,u(-1)\,l(1)\,u(-1)$, with u and l the block-unipotent matrices, has an evident homotopy to I obtained by multiplying every block by T. Apply Lemma 3.4. (2) Write $g\otimes g'=(g\otimes 1)(1\otimes g')$ and apply (1). Up to constant permutation matrices, $g\otimes 1$ is $\mathrm{diag}(g,\dots,g)$, so $[P_{g\otimes 1}]=M'[P_g]$, and likewise $[P_{1\otimes g'}]=M[P_{g'}]$. (3) Write $x=[F]-K$. After adding a common free summand, $F\oplus O^j$ is free of rank M on both pieces. Take g to be the transition between the two bases. □

Proposition 3.6 (correction)**.** *Let V' be a twisted module on A of constant rank whose restrictions to A_a and A_b are good. Then there is an ordinary S of constant positive rank such that $V'\otimes S$ is good.*

Proof. Let $N=\mathrm{rk}\,P(V')$ and $\tau=\rho(P(V'))-1$. Since P commutes with restriction, goodness on A_a and A_b says that $[P(V')]-N$ restricts to torsion classes there, so $\kappa([P(V')]-N)$ restricts to 0 on both for some $\kappa\geq 1$. By Lemma 3.5(3), $\kappa N\tau=[P_g]-M$ for some g, and $\tau^2=0$ by Lemma 3.5(2). Choose e with $R=n\kappa Ne\geq M$ and put $S=P_{g^{-e}}\oplus O^{R-M}$. By Lemma 3.5(1), $[S]=R-e([P_g]-M)$, so $\rho(S)=1-\tau/n$. Hence

$$\rho(P(V'\otimes S))=(1+\tau)(1-\tau/n)^n=(1+\tau)(1-\tau)=1.$$ □

Theorem 3.7. *Let $u_1,\dots,u_k\in A$ be comaximal, and suppose that on each $D(u_i)$ there is a rank-one twisted L_i with $L_i^{\otimes n}\cong\Lambda$. Then there is a twisted module of constant positive rank on* $\mathrm{Spec}\,A$.

Proof. Let Γ be the set of $s \in A$ such that some good twisted module exists on A_s. It contains each u_i, since $P(L_i) \cong O$, and it is closed under multiplication by elements of A. If $s, s' \in \Gamma$, apply Propositions 3.3 and 3.6 over $A_{s+s'}$, where s and s' are comaximal. This gives $s+s' \in \Gamma$. Hence Γ is an ideal and $1 \in \Gamma$. □

The hypothesis $L_i^{\otimes n} \cong \Lambda$ costs nothing: the line bundle $L_i^{\otimes n} \otimes \Lambda^{-1}$ is trivial on a finer Zariski cover.

4. What this changes for Gabber in Unit

(1) Effectiveness (ingredient (2)), Bass cancellation and KO's Theorem 2.11 are not used. Neither are n-th roots beyond the square-zero case: only Lemma 3.1 uses LQ X.5.5.

(2) Theorem 3.7 holds for any ring A. In particular it does not use that A is semilocal or local-global (LQ IX.6) or the symmetric-group trick. The flattening of a T-cover already provides a Zariski cover of the whole tower D on which G is trivial, so KO's 3.4 step, "trivial at one maximal ideal, hence at all of them by S_N", has no counterpart. The route is then: flattening, Theorem 3.7 over D, and push-forward from D to R.

(3) Correction to an earlier remark: LQ X.5.11 (Milnor's theorem) is about Milnor squares, not Zariski squares, so it is not the patching used here. Zariski gluing of modules is LQ XV.4, and in Unit it is just gluing of families.

(4) Inputs to formalise: LQ X.5.5 (via λ_t and γ_t, LQ Problem X-2, using exterior powers from `sag/exterior`); LQ XVI.3.1–3.4 (short); Whitehead's identity; Lemmas 3.2 and 3.5 (elementary); the descent of weight-0 families.